Aalok Thakkar

aalok-thakkar.github.io

Academic Appointment

2024–present — Ashoka University, Sonipat, India
Department of Computer Science
Assistant Professor

Education

2018–2023 — University of Pennsylvania
Ph.D., Computer and Information Science
Thesis: *Example-Guided Synthesis of Relational Queries*
Advisors: Rajeev Alur and Mayur Naik

2015–2018 — Chennai Mathematical Institute
B.Sc. (Honours), Mathematics and Computer Science

Publications

Conference Publications

Hardness of 'Infinite Energy' Puzzles
with Utkarsh Tiwari, Utkarsh Thanvi, Neeldhara Misra, and Armaan Sethi
International Conference on Current Trends in Theory and Practice of Computer Science (SOFSEM 2027)

Bhartrhari's Limitation Theorem
with Rhea
Indian Conference on Logic and its Applications (ICLA 2027)

The Trust a Protocol Needs
with Bhumika Mittal
International Conference on Information Systems Security (ICISS 2026)

Defining Purpose Limited Secrets
with Bhumika Mittal
International Conference on Information Systems Security (ICISS 2026)

Localising Stochasticity in Weighted Automata
with Smayan Agarwal
Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2026)

How Often Does Your Program Fail?
with Arnab Ray
Automated Technology for Verification and Analysis (ATVA 2026). ***Distinguished Paper***

A Forced-Structure Reduction and Verifiable Bounds for Conway's 99-Graph
Conference for AI Scientists (CAISc 2026). ***Best Paper Award***

A Correctness-First Introduction to Computer Science
with Vaani Goenka
Innovation and Technology in Computer Science Education (ITiCSE 2026)

A Good Rubber Duck Does Not Quack
with Ayush Thonge
Innovation and Technology in Computer Science Education (ITiCSE 2026)

Accessibility Beyond Accommodations
with Vaanee Tripathi
ACM India Compute Conference (COMPUTE 2025)

Ancient Algorithms for a Modern Curriculum
ACM India Compute Conference (COMPUTE 2025)

BOOP: Write Right Code
with Vaani Goenka
ACM India Compute Conference (COMPUTE 2025)

SocraticAI: Transforming LLMs into Guided CS Tutors Through Scaffolded Interaction
with Karthik Sunil
ACM India Compute Conference (COMPUTE 2025)

From Transformers to Weighted Automata: Towards Verification of Large Language Models
with Smayan Agarwal, Aslah Ahmad Faizi, and Shobhit Singh
From Data to Models and Back (DATAMOD 2025)

Mobius: Synthesizing Relational Queries with Recursive and Invented Predicates
with Nathaniel Sands, George Petrou, Rajeev Alur, Mayur Naik, and Mukund Raghothaman
Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA 2023)

Relational Query Synthesis $\bowtie$ Decision Tree Learning
with Aaditya Naik, Adam Stein, Rajeev Alur, and Mayur Naik
International Conference on Very Large Data Bases (VLDB 2023)

Example-Guided Synthesis of Relational Queries
with Aaditya Naik, Nathaniel Sands, Mukund Raghothaman, Mayur Naik, and Rajeev Alur
Programming Language Design and Implementation (PLDI 2021)

Additive Collatz Trajectories
with Mrunmay Jagadale
International Conference of The Jangjeon Mathematical Society (ICJMS 2016)

Journal Publications

Reopening businesses and risk of COVID-19 transmission
with Ashley O'Donoghue, Tenzin Dechen, Whitney Pavlova, Michael Boals, Garba Moussa, Manvi Madan, Frank J. DeFalco, and Jennifer P. Stevens
npj Digital Medicine 4, 51 (2021)

Concurrency in Boolean networks
with Thomas Chatain, Stefan Haar, Juraj Kolcak, and Loic Pauleve
Natural Computing 19, 91–109 (2020)

Infinitude of Primes Using Formal Languages
The American Mathematical Monthly 125:8, 745–749 (2018)

Workshop Publications

Complexity of Relational Query Synthesis
with Rajeev Alur and Mayur Naik
11th Workshop on Synthesis (SYNT 2022)

Modular Synthesis of Reactive Systems
with Kedar S. Namjoshi and Richard Trefler
9th Workshop on Synthesis (SYNT 2020)

Verification of Rewriting-based Query Optimizers
with Krishnamurthy Balaji and Piyush Gupta
16th International Workshop on Termination (WST 2018)

Thesis

2023	**Example-Guided Synthesis of Relational Queries** Ph.D. thesis, University of Pennsylvania.

Teaching

Monsoon 2026	CS-2212-1: Data Structures and Algorithms
Summer 2026	FC-0306: Quantitative Reasoning and Mathematical Thinking
Spring 2026	CS-3330: Theory of Computation CS-3610: Information Security
Monsoon 2025	CS-1102: Introduction to Computer Science CS-5310: Symbolic Logic and Applications CS-6101: Computer Science Research Methods and Seminars
Summer 2025	FC-0306: Quantitative Reasoning and Mathematical Thinking CS-1102: Introduction to Computer Science
Spring 2025	CS-1102: Introduction to Computer Science CS-5440: Trustworthy AI

Monsoon 2024 CS-5310: Symbolic Logic and Applications

Grants and Fellowships

2026–2029 VSAC Young Faculty Grant, Ashoka University

2026–2027 INSA Visiting Scientist Programme
Visiting the Chennai Mathematical Institute to work with Aiswarya on the synthesis of string transducers.

2026– ANRF Prime Minister Early Career Research Grant
Robust and scalable program synthesis.

2025–2026 Data & AI Adoption Fellowship, EkStep Foundation
AI Safety Benchmarks for Education and Agriculture.

2025–2026 Darjeeling Revival Fellowship, Alt Carbon
Geostatistical modelling for enhanced rock weathering.

2025–2026 Adaptive and Inclusive Pedagogy Grant, Mphasis
Curriculum design for an adaptive and inclusive classroom.

2026 VSAC Transdisciplinary Research Grant, Ashoka University
Two projects: Circle of Joy (with Devapriya Roy and Annie Baxi) and Animal Acoustics (with Meghna Agarwala and Nandini Chatterjee Singh).

2026 Nagarathna Memorial Grant
Translation of *Nayi Kavita* poets into English.

Professional Service

Organiser Verification, Logic, and Cryptography (VeLCro) 2026, CMI.
ACM COMPUTE 2026, Amrita Vishwa Vidyapeetham.
FSTTCS 2026, IIT Delhi.
Formal Methods Update Meeting 2026, Krea University.
ACM COMPUTE Regional 2026 (North India), Ashoka University.
IndiCS Seminar 2: Automated Synthesis, 2025, Infosys Mysore (coordinator).
FSTTCS 2024, FSTTCS 2025.
IARCS Verification Seminar Series.
Logic Society of Delhi.

Editor Corresponding editor, *Bhāvanā* magazine.

Reviewer Programme committees and review activities for ICLA 2026, MFCS 2026, OOPSLA 2026, ATVA AEC 2025, PLDI SRC 2025, ISEC 2025, POPL AEC 2022, *ACM Computing Surveys*, and the *Indian Journal of History of Science*.

Memberships	Association for Computing Machinery (ACM India). Vijnana Bharati.